What we tested

Does redesign-only training broaden contact predictions and reduce their dependence on available evolutionary information?

We compared the exp364 MPNN-only 1.5B checkpoint at step 186884 with the exp277 full-data default at step 266344. The primary cohort is 97 natural eval-val proteins; 19 designs are reported separately. Eval-test was not scored.

The frozen design has 300 fixed-prompt samples plus 100 matched randomized prompts per model/protein, at temperatures 0.7, 1.0 and 1.3. All 278,400 samples are saved. Temperature 1 is primary. Intervals bootstrap proteins, preserving model pairing, with 10,000 draws.

Diversity increases after the controls

On natural proteins, mean fixed-prompt Jaccard distance rises from 0.708 to 0.813. The paired difference is +0.1046, with 95% CI [0.0866, 0.1229].

Matching every paired sample's contact count gives +0.1064 [0.0891, 0.1244]. Randomized prompts give +0.1008 [0.0836, 0.1189]. Long-range contacts show the same effect. Independent seed-block differences are +0.1050, +0.1042 and +0.1046.

MPNN-only generates modestly more contacts, but the long-range fraction is similar. Count, prompt and seed-block controls support a real difference in the contact-sampling distribution.

Extra diversity mainly adds benchmark errors

Single-rollout precision falls from 0.411 to 0.298. After 300 fixed-prompt samples, true-contact coverage is essentially tied: 0.964 versus 0.967, paired difference +0.0029 [−0.0033, 0.0095].

Distinct false contacts rise from 5,642 to 8,902 per protein: 58% more, with a paired increase of 3,260 [2,367, 4,298]. Fixed-prompt ensemble R-precision falls from 0.539 to 0.454.

The standard 100-randomized-prompt R-precision is 0.554 versus 0.463, difference −0.0911 [−0.1113, −0.0719]. Default ensemble accuracy remains higher throughout the temperature sweep. Increased diversity does not establish useful alternatives or physical conformations.

Neff gives contrast-specific support

Neff is redundancy-weighted depth at 80% identity, not raw MSA sequence count. Fixed tiers contain 4, 10, 28 and 55 natural proteins.

The all-range high-minus-low accuracy drop is 0.399 for default and 0.329 for MPNN-only: difference −0.0705 [−0.1026, −0.0401]. However, absolute low-Neff accuracy is 0.280 versus 0.263; MPNN-only does not improve it.

The continuous interaction is inconclusive: −0.0080 [−0.0276, 0.0165] per Neff decade. Long range, length/viral adjustment and nonviral-only slopes are also inconclusive. The largest accuracy gap occurs in the middle tiers. Four lowest-tier proteins limit generalization. No natural protein has raw MSA depth below 10.

Patterns and mechanisms

Both models have contact dependencies beyond sample-size and contact-frequency margins. Excess Jaccard SD over exact-margin shuffled controls is 0.0197 for default and 0.0169 for MPNN-only. Half/full-chain differences are small relative to that excess. This does not identify physical states or uniquely stronger MPNN-only coordination.

Eight contact-map examples were chosen by Neff rank before inference. Some retain similar recurring patterns; others show diffuse MPNN-only frequencies and lower accuracy.

Existing sequence searches cover native AFDB/ESM-Atlas. The actual 162.6-million-document redesign corpus has no complete available sequence-search or KNN audit. Removing native documents retains structural ancestry and cannot establish an absence of retrieval or memorization.

Secondary designs and execution checks

For 19 designs, the Jaccard difference is +0.0457 [−0.0017, 0.0976], inconclusive and smaller after matching counts. Randomized-prompt all-range R-precision is 0.708 versus 0.661, difference −0.0469 [−0.0857, −0.0159]. Designs are never pooled with natural proteins.

All 12 H100 production shards succeeded. Retained conditions total 5.20 GPU-hours of inference, excluding setup and retries. All malformed and capped outputs remain included: 1,501 continuations and 17 final caps, with no continuation at temperature 1.

Validation covers 1,392 conditions, 27,840 metric rows, exact prompt/seed pairing, continuation prefixes, context limits and checkpoint hashes. Ten unit tests pass. Upload recovery changed publication only; the scientific worker/parser programs were identical.

Interpretation and reproducibility

Greater diversity is supported on natural proteins, but it mostly adds false-contact discovery and lowers ensemble accuracy. The Neff hypothesis has partial support from the extreme-tier contrast, not a clear continuous interaction or better absolute low-Neff prediction. The full-data default remains stronger here.

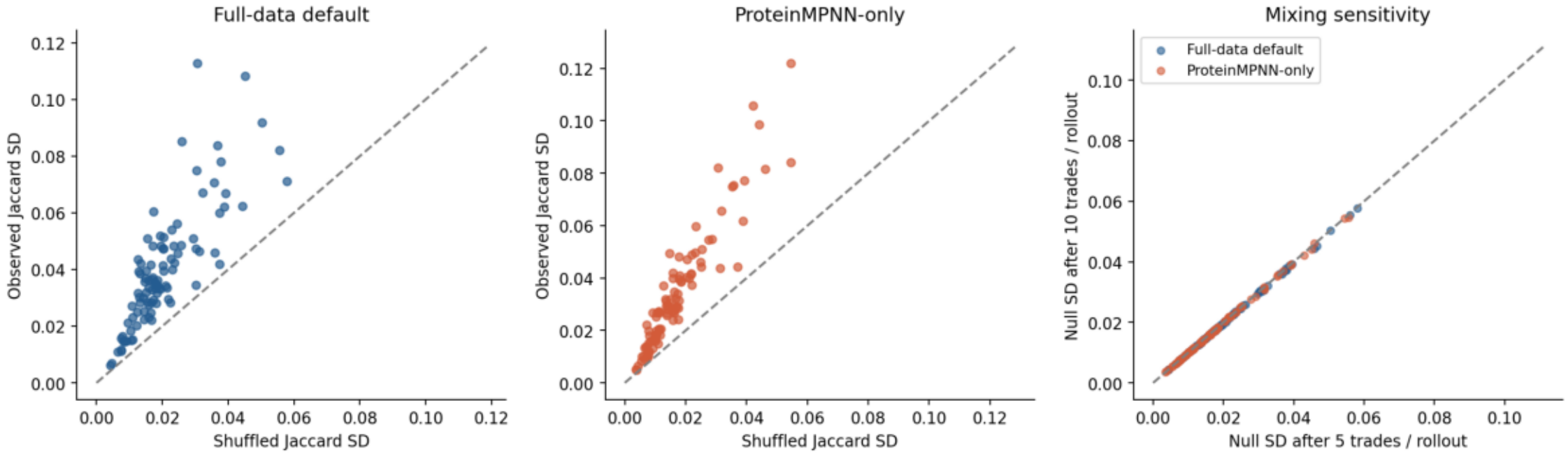
One selected checkpoint per recipe, different training exposure/schedules, observational Neff associations, sparse low-Neff data and incomplete redesign-sequence searches limit causal interpretation.

The README is the complete report. Figures regenerate from committed small tables; 921 MB of compressed raw samples and all inputs are public, with checksums and anonymous-read verification. Each appendix figure prints its generating command.

Report and source: https://github.com/Open-Athena/MarinFold/pull/374 Public bucket prefix: open-athena/MarinFold/data/exp373-mpnn-diversity-neff/v1

coordination_control.png

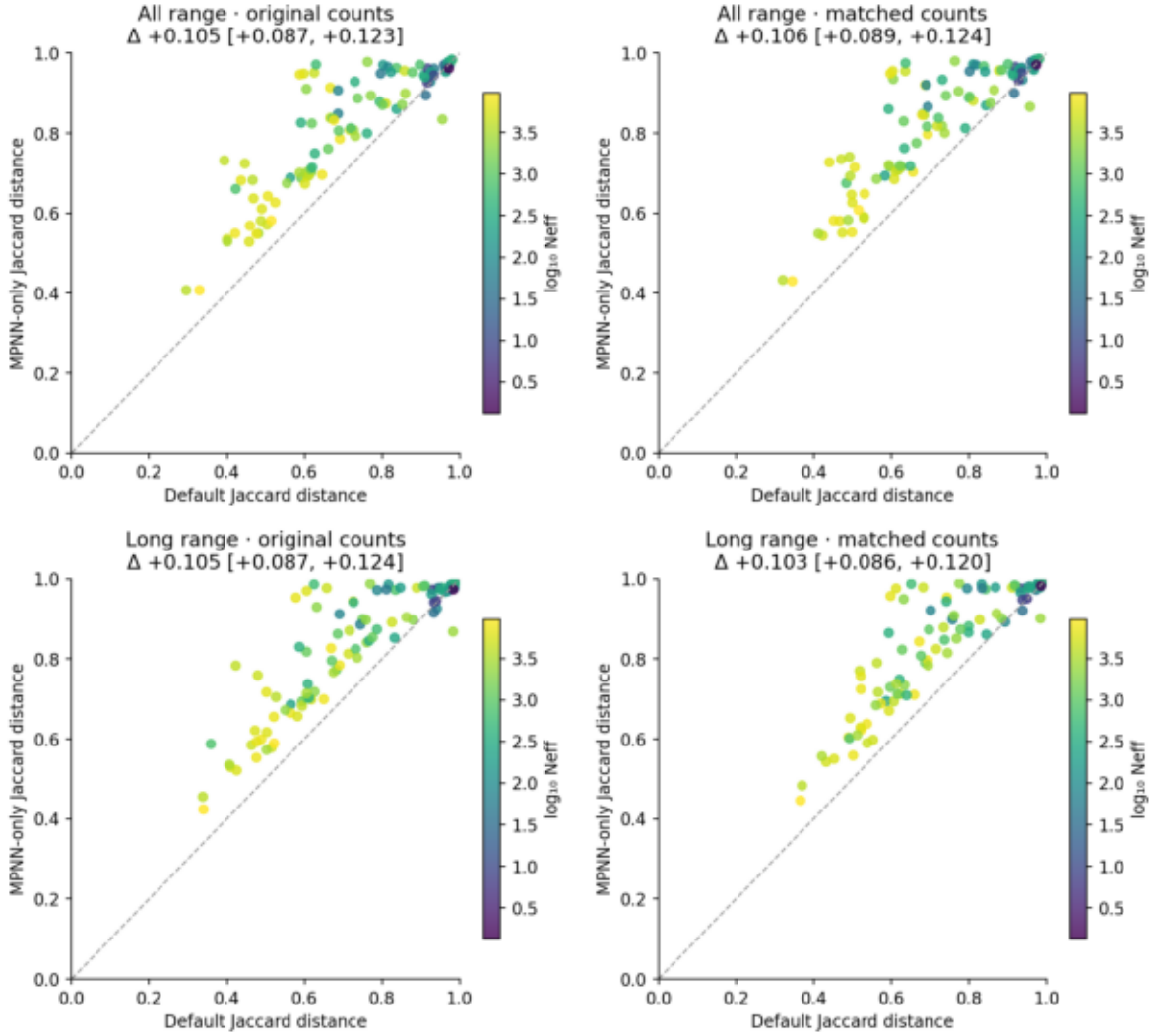
Fixed prompts, T=1. Curveball shuffles preserve every sample size and contact frequency. Ten null replicates; independent half/full chains assess mixing sensitivity. Coordination does not establish conformations.



$ plot.py

diversity.png

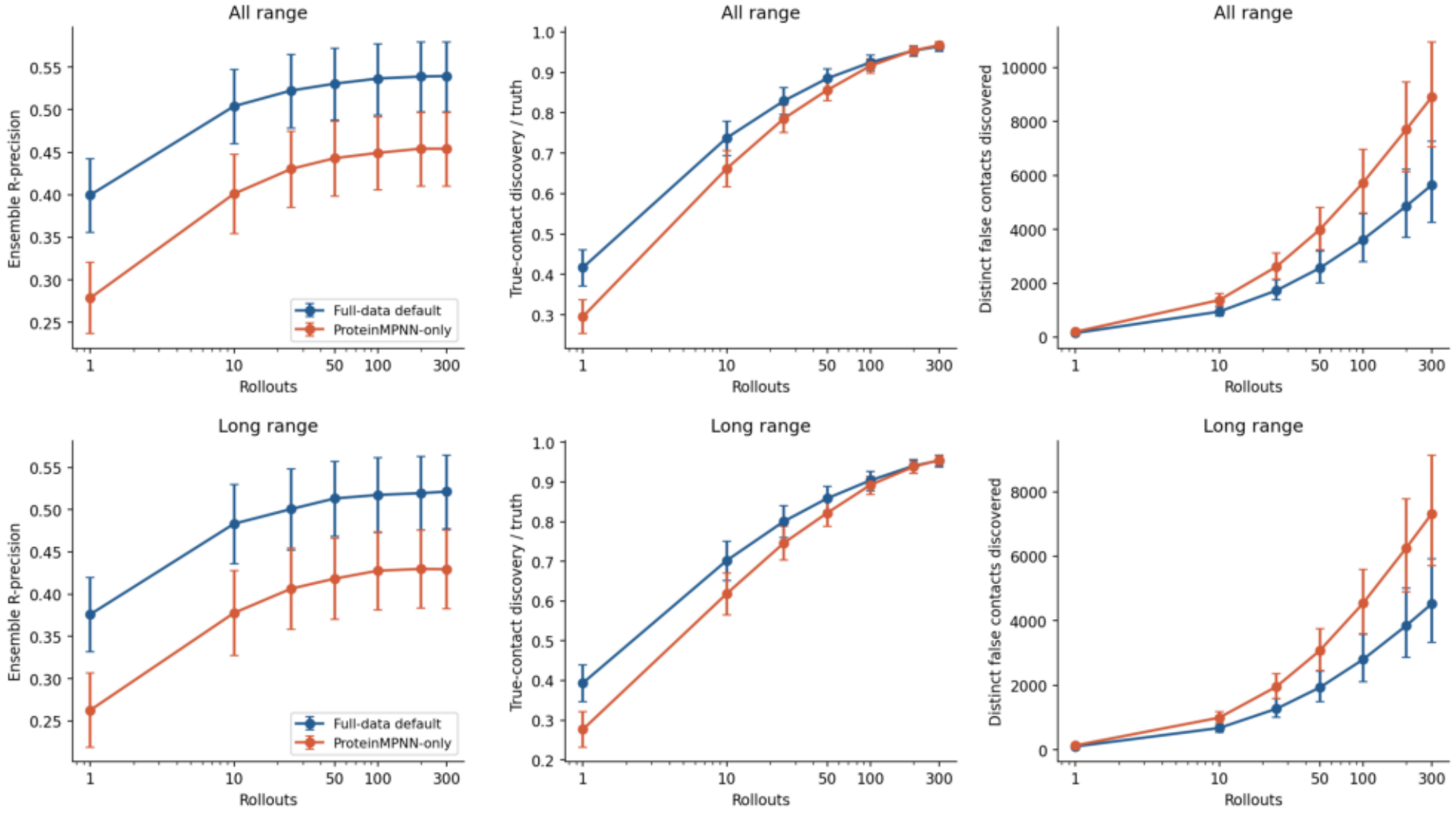
97 natural proteins, T=1, 300 fixed-prompt samples. Points are proteins; Δ is MPNN-only minus default (paired 95% CI). Count control thins paired samples to equal contact counts.



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ensemble_discovery.png

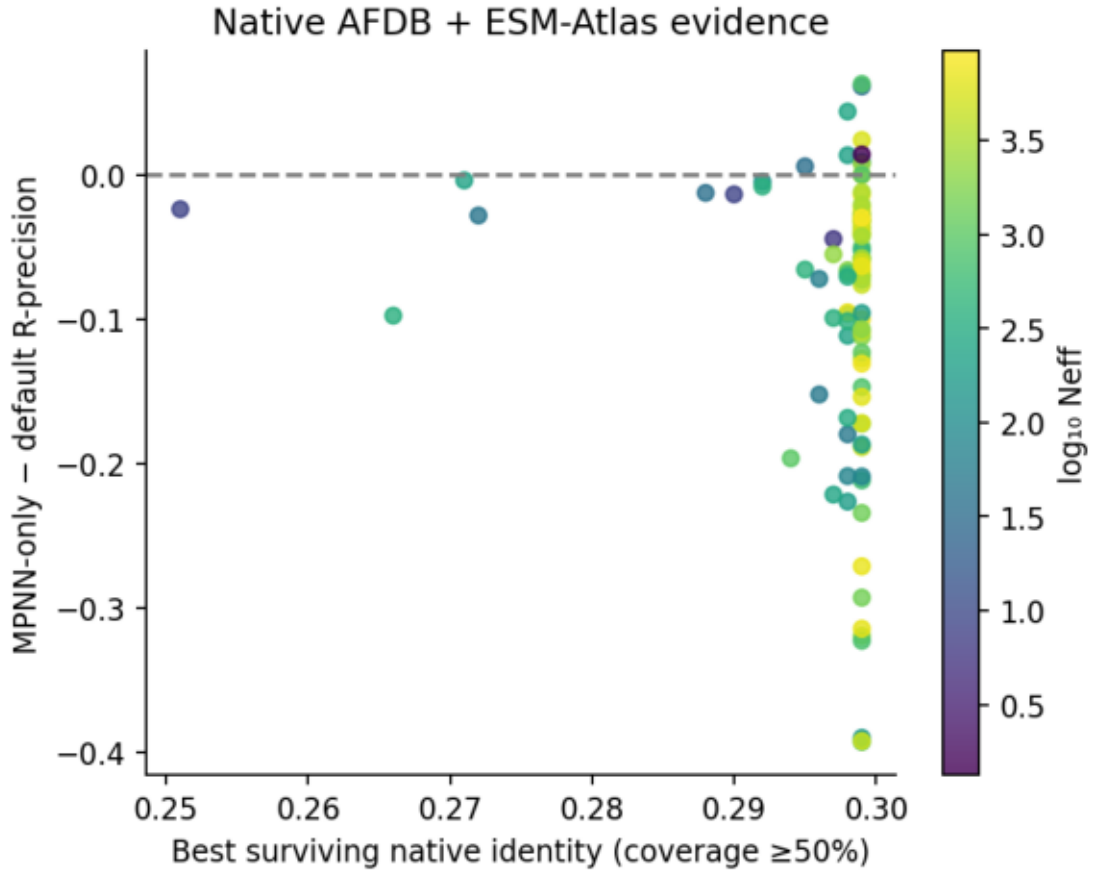
Natural proteins, T=1, fixed prompts. True-contact coverage is shown alongside false discovery and ensemble accuracy. Error bars: 95% protein-bootstrap intervals.

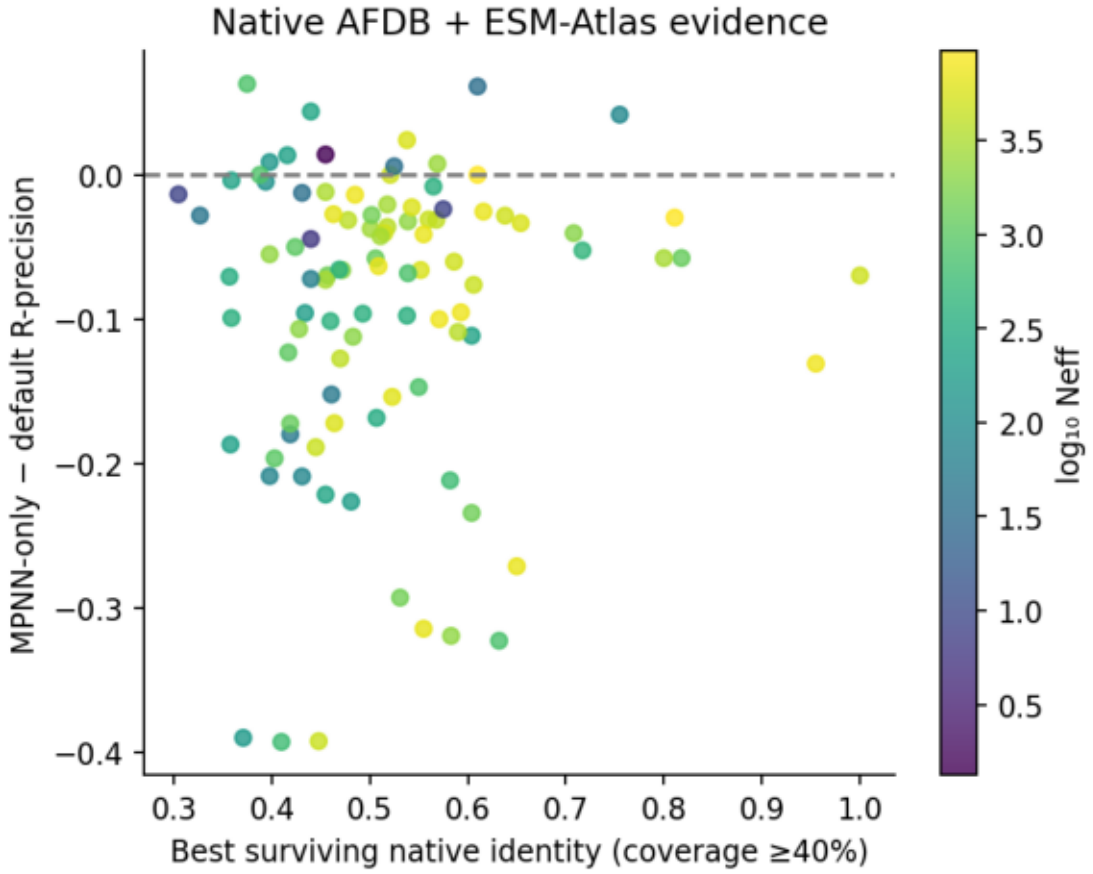



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native_similarity_audit.png

Exploratory exp245 native AFDB/ESM-Atlas alignment evidence. The 50% coverage filter restricts identity; 40% exposes local matches. Redesign sequences were not searched.

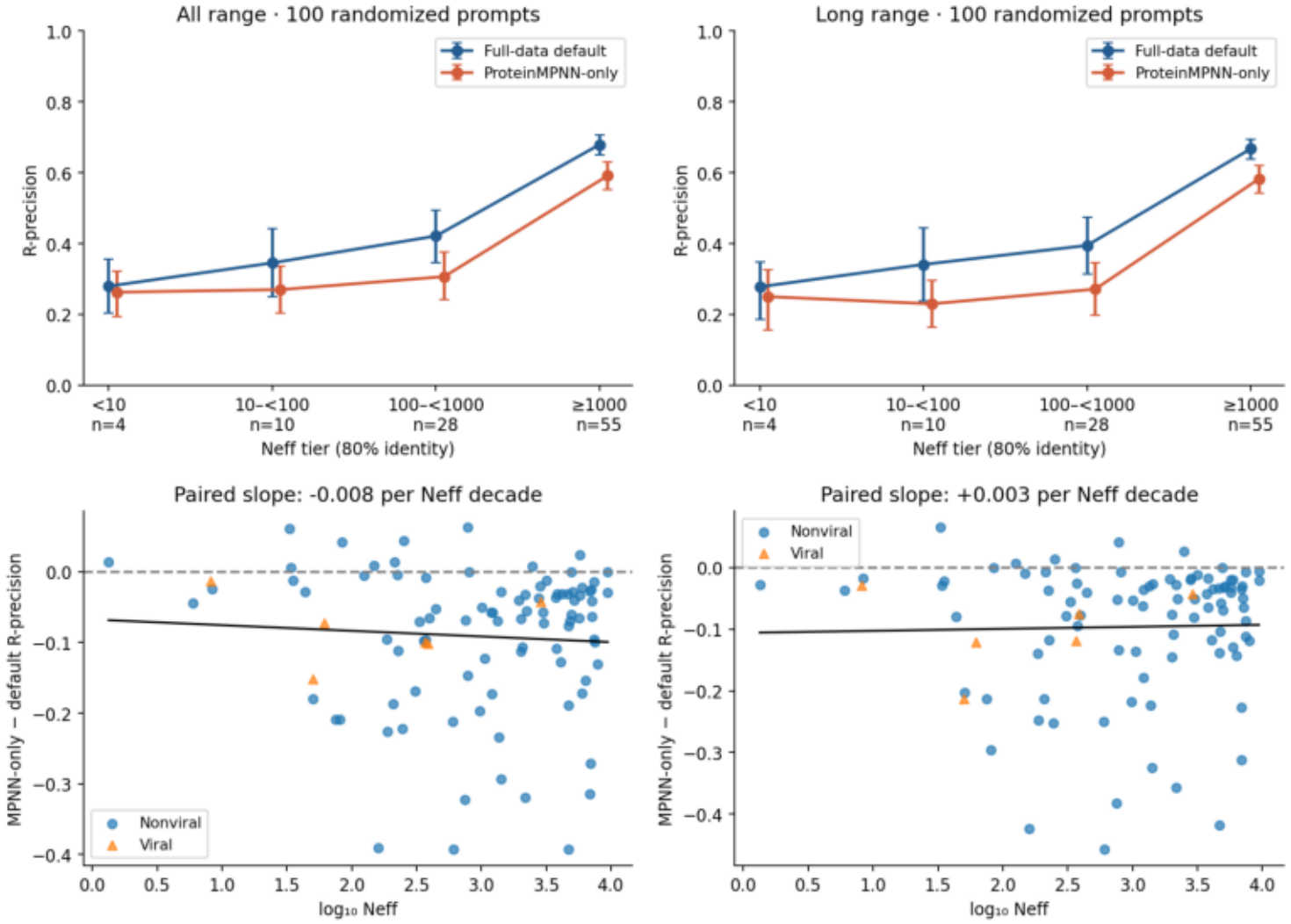




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neff_dependence.png

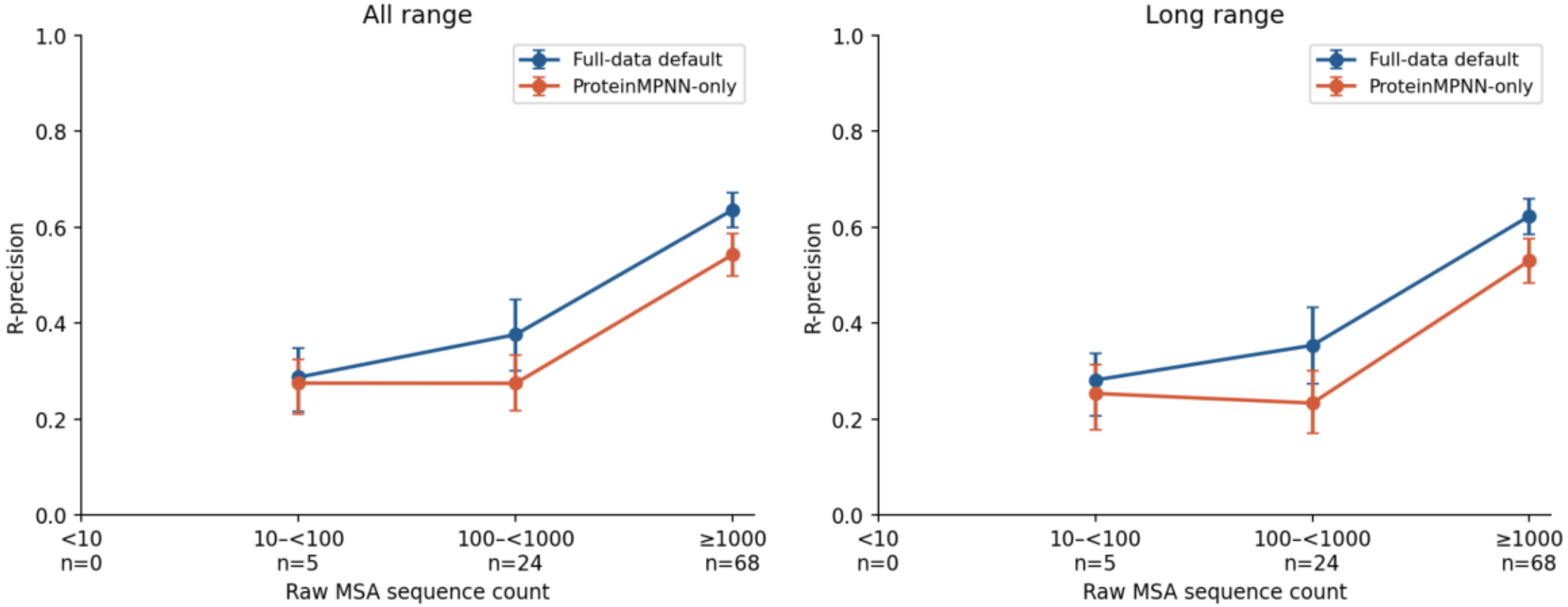
Primary comparison: 97 natural proteins, T=1, 100 randomized prompts. Top: fixed tiers and bootstrap 95% CIs. Bottom: paired differences and unadjusted OLS.



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raw_depth_sensitivity.png

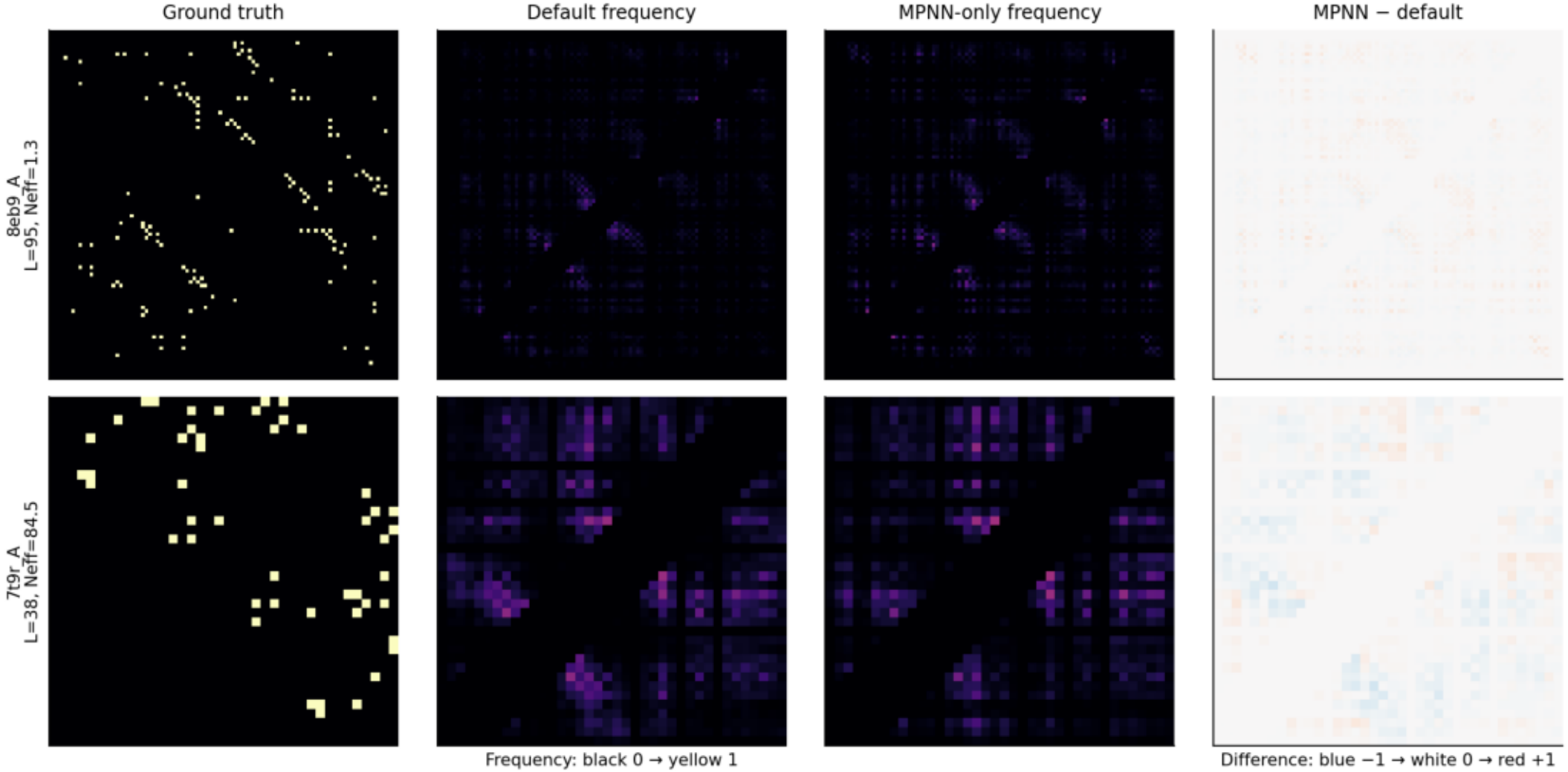
Raw sequence-count sensitivity on the same 97 proteins and primary condition. Empty tiers are explicit; raw depth is distinct from Neff.



$ plot.py

representative_contact_maps_1.png

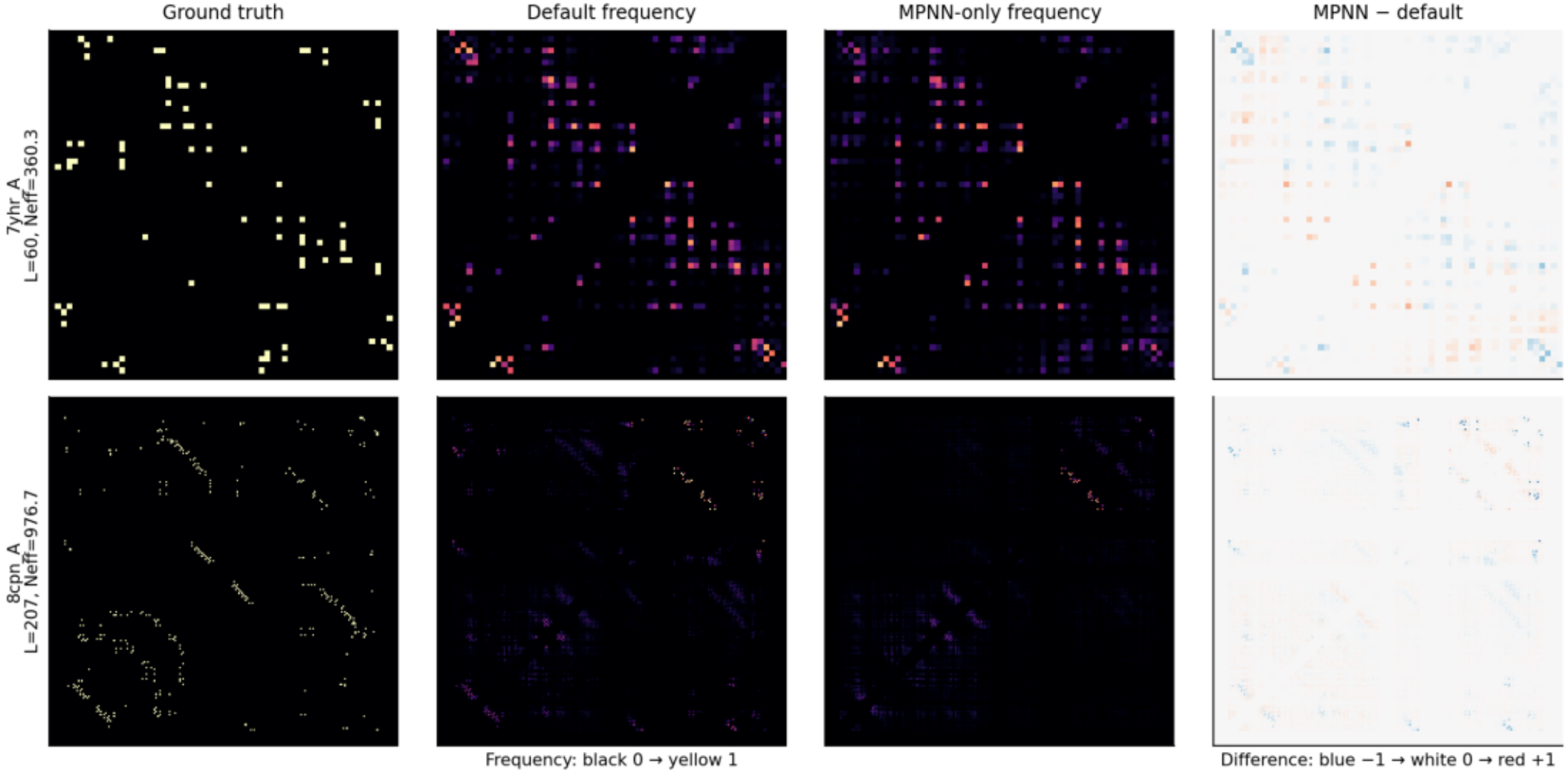
Eight proteins preselected by Neff rank; two shown per page. Resolved-pair truth and frequencies from 300 fixed-prompt samples, T=1. Frequencies use 0–1 and signed differences −1 to +1.



$ plot.py

representative_contact_maps_2.png

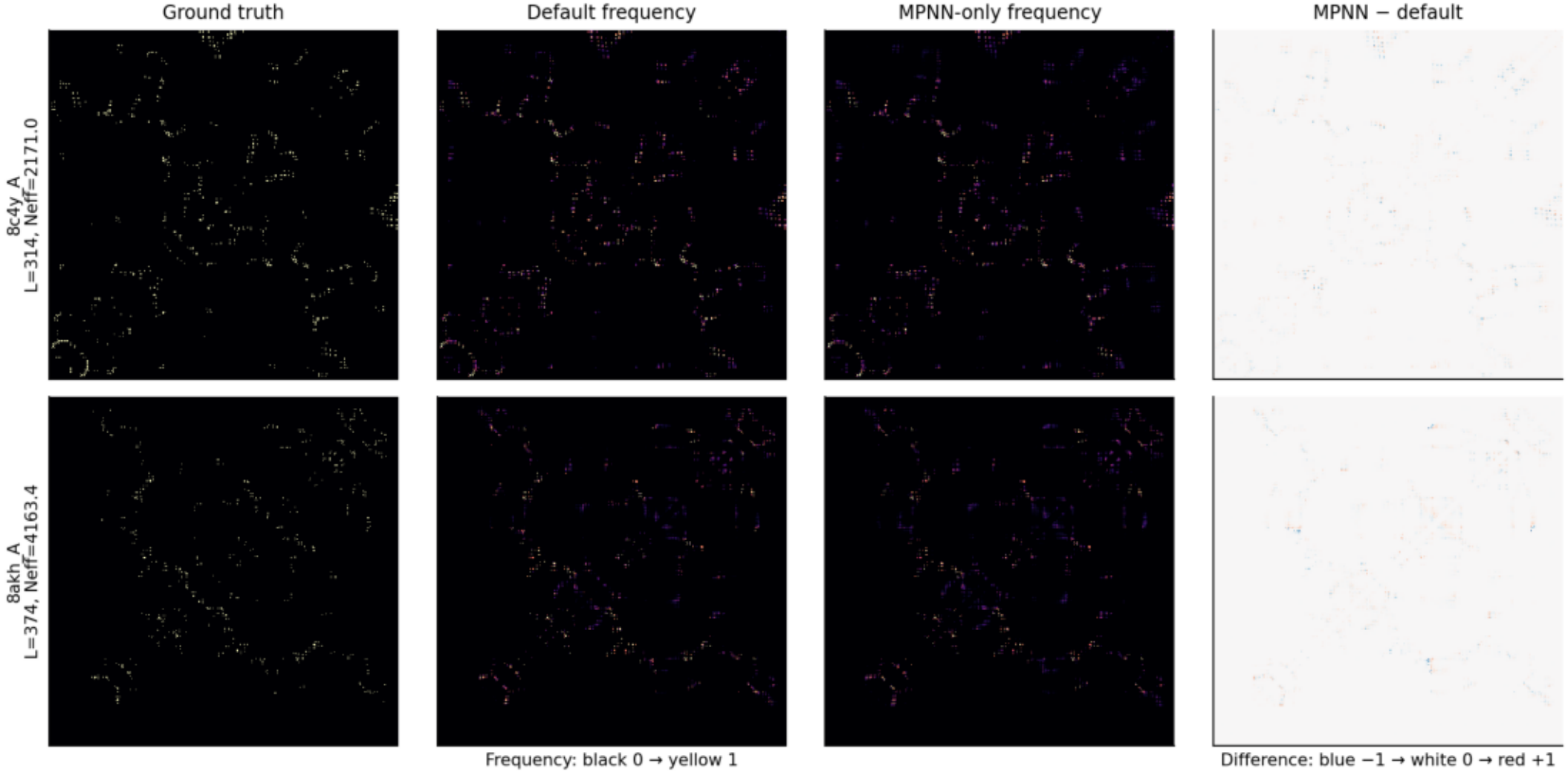
Eight proteins preselected by Neff rank; two shown per page. Resolved-pair truth and frequencies from 300 fixed-prompt samples, T=1. Frequencies use 0–1 and signed differences −1 to +1.



$ plot.py

representative_contact_maps_3.png

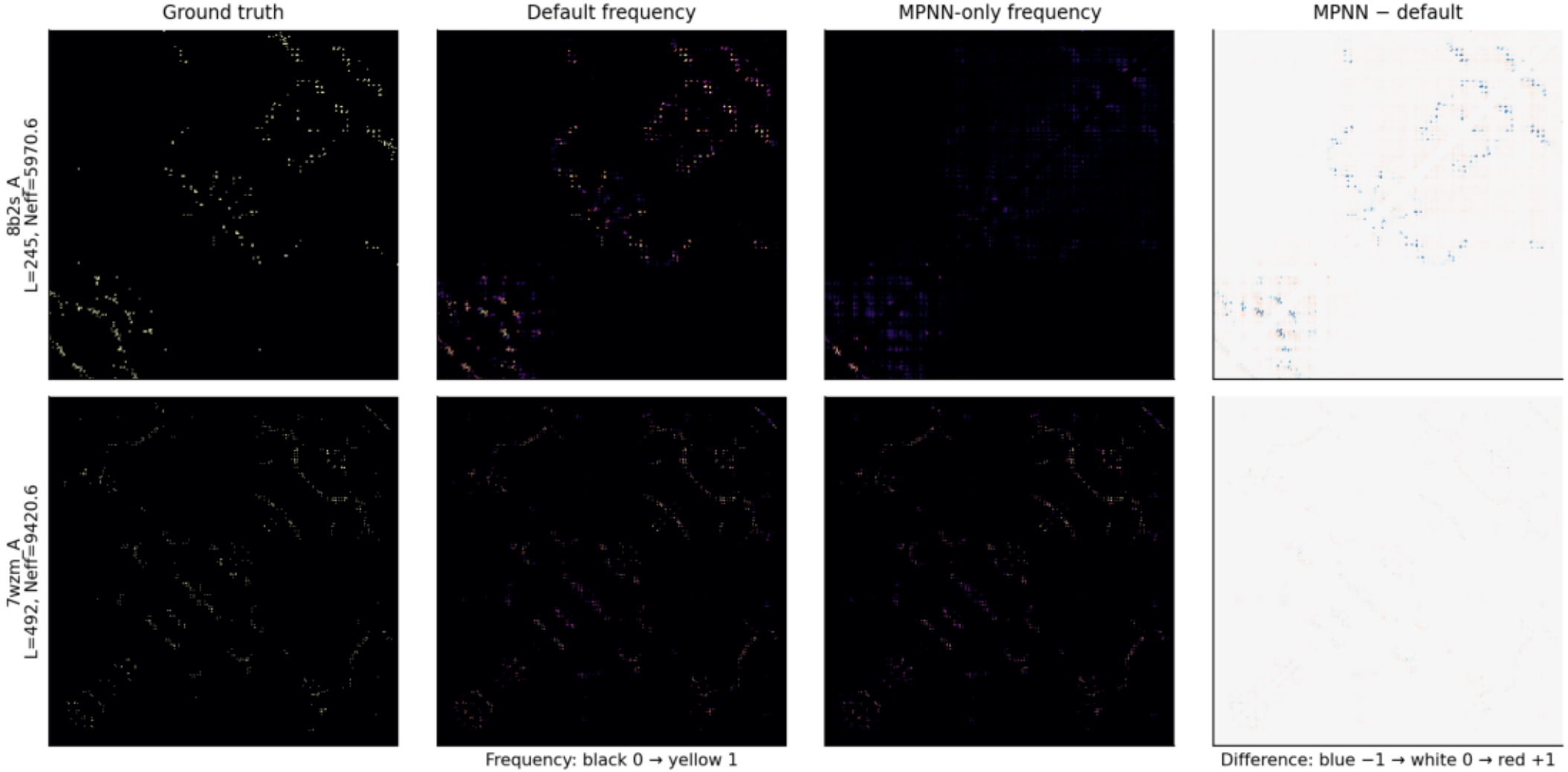
Eight proteins preselected by Neff rank; two shown per page. Resolved-pair truth and frequencies from 300 fixed-prompt samples, T=1. Frequencies use 0–1 and signed differences −1 to +1.



$ plot.py

representative_contact_maps_4.png

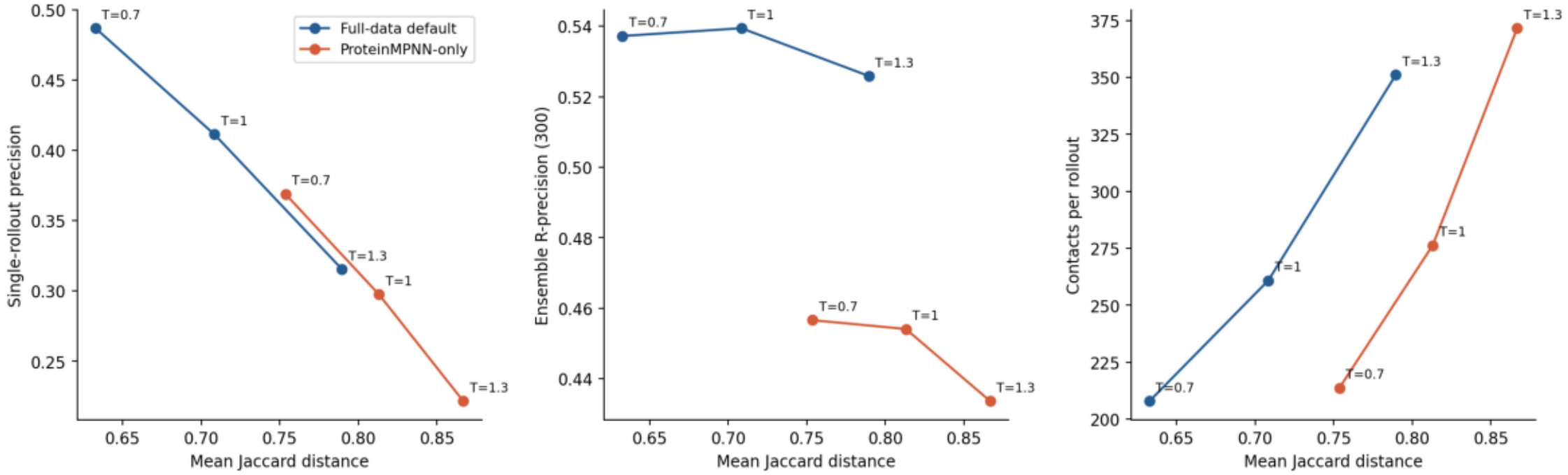
Eight proteins preselected by Neff rank; two shown per page. Resolved-pair truth and frequencies from 300 fixed-prompt samples, T=1. Frequencies use 0–1 and signed differences −1 to +1.



$ plot.py

temperature_tradeoff.png

Natural proteins, fixed prompts. Means at temperatures 0.7, 1.0 and 1.3 show the tradeoff between diversity, precision, ensemble accuracy and output size. Descriptive comparison; no temperature was selected.



$ plot.py